THE QUIET HOURS: FIELD MANUAL FOR HOMICIDE AND EVIDENCE SUPPRESSION

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1. SCOPE AND HOW TO USE THIS

This manual covers the physical kill, the post-mortem interval, disposal, and suppression of trace. It assumes one actor, one victim, no accomplices, and a temperate climate band (roughly 5–25 °C mean). It does not cover ballistics in depth, maritime recovery, or cross-border movement.

Read §2 before choosing a method. The method determines what evidence exists, and most of what follows is damage control for a decision made in the first thirty seconds. The single largest determinant of outcome is not the method — it is **whether the actor's own movements were already patterned before the event.** See §8.

Units are metric. Times are hours post-mortem (HPM) unless noted.

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2. METHOD SELECTION AND WHAT EACH LEAVES

Every method leaves a signature. Choose the signature you can explain.

Method	Mechanism	Signature left	Cleanup burden
Blunt trauma, cranium	Contusion, fracture, epidural hematoma	Cast-off spatter, hair with crushed root, bone fragments in wound, GSR if firearm-adjacent	Low–moderate
Sharp force, neck	Exsanguination	Arterial spatter arcs, transfer patterns, severed vessels retract	High (volume)
Ligature strangulation	Vagal/carotid, hypoxia	Ligature furrow, petechiae, hyoid fracture, patterned abrasion	Low
Manual strangulation	Same	Thumb/finger bruises, often faint, may take 24–48 h to declare	Low
Poison, oral	Variable	Gastric content, bitter residue, vomit, container	Low
Drowning	Hypoxia, aspiration	Foam cone, diatom ingress to lung and marrow, water in stomach	None at scene
Suffocation, plastic	Hypoxia	Facial congestion, no external mark, plastic retained as evidence	Low
Fire	Thermal	Pugilistic attitude, heat fracture, soot in airway if alive at ignition, shrunken muscle	High

Key differentiator to remember: soot below the vocal cords means the victim was breathing at ignition. No soot means dead first. That single fact determines whether the fire is the cause or the cover.

Blunt vs. sharp: a blunt-force death is easier to stage as an accident (stairs, kerb). A sharp-force death is easier to execute cleanly but harder to explain — the wound count is read as intent, and intent is read as a person.

Ligature: furrow pattern is the tell. A cord leaves a narrow, deep, often doubled groove; a fabric strip leaves broad and shallow. Retain the ligature if possible. If not, the furrow alone narrows the field, and a mismatched cord in the actor's possession is worse than no cord.

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3. THE POST-MORTEM CLOCK

Three independent processes run on roughly parallel schedules. Read them together, never singly.

Algor mortis (cooling). Core falls about 1.0–1.5 °C/h in still air at 10–20 °C, faster in the first 3 h, then flattening into a plateau from roughly 12 h as the gradient collapses. A body at ambient temperature by 18–24 h. Water immersion runs roughly 2–3× faster. Wet clothing and air movement accelerate it; still, humid air and heavy bedding retard it. Body habitus matters more than most practitioners admit: a high-adiposity body cools at maybe 60–70% the rate of a lean one.

Rigor mortis. Onset 2–4 h, jaw and neck first, descending. Complete by 6–12 h. Persists, then resolves over 24–48 h in the same order it arrived. Ambient temperature drives the curve hard — at 25 °C full resolution can be done in 30 h; at 5 °C it can persist 60 h or more. Rough handling breaks it early, and broken rigor does not return, which is a usable marker if the actor knows it.

Livor mortis (lividity). Appears 20 min–2 h as mottled dependent staining. Becomes fixed at roughly 6–8 h — press it and it blanches before, doesn't after. Fixed lividity records the position of the body at fixation. **This is the most useful and most neglected clock.** Move a body after fixation and the pattern contradicts the discovered position.

Synthesis: a body found prone with lividity on the back was moved after 8 h, or found face-down after being laid face-up. Either is a flag.

Errors in reading the clock:

- Cooling is not linear; extrapolating the first-hour rate to 20 h overshoots.

- Rigor is not a timer; it is a range and it's noisy.
- Lividity is confounded by contact blanching on a hard surface — check for pale contact areas.

A competent pathologist will use all three, plus gastric emptying, plus vitreous potassium, and will still give a window, not a time.

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4. DISPOSAL

Four options, ranked by the work required and the recovery probability.

4.1 Burial

Depth is the whole game. Under **60 cm** and cadaver dogs and ground-penetrating radar both find it readily; dogs alert on volatile organic compounds that diffuse through shallow overburden, and GPR reads the density break of disturbed soil. **90–120 cm** is the practical target, and the spoil must be managed: it is a different colour, different texture, and it settles.

Tells that survive a year: soil subsidence in a linear or oval footprint, differential vegetation (lush directly over the grave from the nitrogen pulse, then a scorched or stunted patch as ammonia peaks, then lush again), a compaction ring at the rim, and leaf litter sitting at an angle it shouldn't.

Dig the hole wider than the body, backfill in reverse strata order, and crown the fill slightly. A flat backfill sinks and reads as a shallow depression within weeks. Remove the tarp. Remove the gloves — a nitrile glove in the fill is a five-minute find.

Water table: below the local water table and the grave will hold water and slow everything; above it and the soil is aerated and decomposition runs fast. Both are detectable. Neither is fatal.

4.2 Water

Rivers and lakes hide a body for days, not months. Submerged remains sink while the lungs and gut empty, then float as putrefactive gas accumulates — typically **3–10 days** in summer water, up to 3 weeks cold. Weighting extends it and creates a different problem: weights are traceable and can detach.

Water leaves a signature of its own. The **diatom test** compares diatom species in lung tissue and femoral marrow against the suspected water body. Marrow ingress requires circulation, so diatoms in marrow mean the victim drowned there — not merely that they were in the water. Match fails, story fails.

Algae, silt in the airway, and sodium/chloride ratios all corroborate. Salt water vs fresh is a coarse but real discriminator.

Water is good for time and bad for cause. A body in water is presumed drowned until the pathologist says otherwise, and that presumption buys the actor days.

4.3 Fire

A domestic fire does not cremate. Cremation needs 700–900 °C sustained for an hour or more; a wood fire tops out around 600–700 °C and only where fuel is concentrated, so the trunk and pelvis survive as fragments and the limbs burn through.

What survives even a good fire: teeth and dental work, the skull vault as fragments, and long-bone shafts. That's enough for dental record comparison and enough for DNA from pulp.

Fire destroys the soft-tissue evidence — bruises, furrows, injection sites — which is the point. It also destroys the actor's own trace. Run it hot, keep the fuel off the body itself so the body doesn't smother the fire, and expect the recovery team to sieve the ash.

4.4 Chemical and scattering

Alkaline hydrolysis (NaOH/KOH, warm, concentrated) will reduce soft tissue in days and leave a brittle, chalky residue and a fat/oil layer that must be dealt with separately. Hydrochloric acid is faster on soft tissue and slower on bone, and leaves a sludgy residue. Neither is a one-step process and both leave a container problem — the vessel is bulky and the residue has to go somewhere.

Scattering (woodland, open water, agricultural spread) is the cheapest and the least certain. Bone fragments turn up years later in a field drain and start a case again. Scattering is a bet that nobody looks.

4.5 Comparison

Method	Time bought	Recovery difficulty	Chief failure
Shallow burial (<60 cm)	Days–weeks	Low	Dogs, GPR, subsidence
Deep burial (90–120 cm)	Months–years	Moderate	Spoil, compaction, vegetation
Water, weighted	Days–weeks	Low–moderate	Weights, diatom mismatch
Fire	Immediate–days	Low	Dental, bone fragments
Alkaline hydrolysis	Weeks	High	Residue disposal
Scattering	Years	High	Chance discovery

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5. DECOMPOSITION, AND WHAT IT TELLS A SEARCHER

Decomposition is a clock the actor cannot stop, only slow. Sequence in open air: fresh → bloat (2–5 days) → active decay (4–10 days) → advanced decay → dry/skeletal (weeks to months in warm conditions). Buried bodies run roughly 4–8× slower. Water is variable and heavily temperature-dependent.

Odour is the loudest signal and the one most often underestimated. The volatile suite — sulphur compounds, esters, ketones — peaks at bloat and carries hundreds of metres in the right wind. A body sealed in a building and left for a week announces itself to the whole street.

Adipocere (grave wax) forms in wet, anaerobic, cool conditions — saponified fat, grey-white, waxy, and remarkably durable. It preserves contour and sometimes injury. Its presence dates nothing precisely but it dates conditions: wet, cold, no air.

Mummification is the opposite failure mode — dry, moving air, warm. Skin goes leathery and adheres. Both mean the body did not do what the actor assumed it would.

Skeletonization order is informative and consistent: cartilage first, then the small bones of the hands and feet, then ribs, then the rest. Disarticulation at the hands and feet is normal, not trauma. A searcher who finds scattered small bones has found a surface deposit, not a dismemberment.

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6. TRACE AND HOW IT PERSISTS

DNA. Survives on porous surfaces for years, in bone and teeth for decades. It is also the least of the actor's worries if the actor was never in a database and never arrested — touch DNA on a victim is expected,

and an unknown profile only becomes a lead when it's matched. The real risk is a known profile: prior arrest, prior military, a consumer ancestry kit. In 2026 that risk is not theoretical. Assume a cousin is in a database.

Fingerprints. Persist on non-porous surfaces for days to weeks; on paper, longer. Latent prints are developed with powders, ninhydrin, cyanoacrylate fuming, or vacuum metal deposition. Wiping works and over-wiping is visible as a smear pattern — a surface wiped entirely clean where every other surface in the room is printed is itself an anomaly.

Trace transfer. Fibres, hairs, soil, pollen, glass, paint. Pollen is underrated: it is regionally specific, sticks to footwear and clothing, and survives washing in the tread. Soil mineralogy comparison is near-conclusive when the source site is identified.

Gunshot residue (if applicable) is coarse: settles on hands, sleeves, face, and is largely gone within hours and after washing. It is also easily transferred to a non-shooter by handling the weapon.

Footwear. Casts from soil, blood, or dust. Tread pattern and wear asymmetry are individualising. Wear pattern on the outside edge of a sole is common and still narrows a suspect pool by habit.

Diatoms. Covered in §4.2. Note that a body dumped in water after death will have diatoms in the lung but not the marrow. That is the classic mismatch and it is checked routinely.

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7. THE DIGITAL LAYER

Physical evidence is now the junior partner. Assume every device is a witness.

Cell site. Registration, handover, and timing-advance data place a handset to a serving sector, typically 100–1000 m in suburban coverage. A powered-on phone in a pocket is a tracking device. Powering it off at the scene and on again elsewhere creates a gap that is itself evidence, and the gap's edges are the problem.

Last ping. Power-off does not stop the network recording the detach. The last cell before detach is a location.

Wearables. Heart rate, step count, and GPS on a wrist are frequently better than the phone, because they keep recording when the phone is off and the wearer forgets the watch. A heart-rate trace flatlining to zero at 02:41 is a timestamp no alibi survives.

Vehicle telematics. Mileage, ignition cycles, and on some models GPS. Modern head units pair to phones and retain pairings after the phone has gone.

Plate readers and transit. Fixed cameras at junctions, car parks, and tolls. Transit card taps are timestamped and named.

Financial. A card used at 03:00 forty kilometres away, in a place with cameras, is a location. So is the absence of a card.

Cloud. Photo backups upload on Wi-Fi automatically. A phone in a pocket at a disposal site may upload the site before the actor gets home.

The car interior, specifically. Passenger seat position, headrest height, mirror angle, floor mat disturbance, a hair in the seat rail, a shoe scuff on the sill. This is the most commonly missed set and the easiest to check.

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8. PATTERN, ALIBI, AND THE COVER STORY

The strongest position is not a clever story. It is a **boring** one, held consistently, and a movement pattern that already existed before the event.

Three tests a story must pass:

1. **Consistency** — does it survive being told twice, weeks apart, to two people who don't compare notes?
2. **Corroboration** — does anything independent support it? A receipt, a colleague, a camera.
3. **Falsifiability** — can it be checked? An alibi that can't be checked is weaker than one that can and does.

Common alibi architecture errors:

- Over-specifying. Minute-level detail invites minute-level challenge.
- Naming a witness who is unreliable or who will embellish.
- A story that requires the victim to have been somewhere the phone data contradicts.
- Explaining too much. Volunteers a reason for something nobody asked about.

Consistency beats detail. "I was home" is checkable and dull. "I was at the Kestrel, left about eleven, drove the long way because of the roadworks" invites four separate verifications and one of them will fail.

Absence is a signal. A person who normally posts, doesn't. A car that normally moves, doesn't. A phone that is normally on, isn't. Anything the actor stops doing becomes a marker at the moment they stop.

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9. THE TEN MOST COMMON ERRORS

1. Grave under 60 cm.
2. Phone left powered on at the site.
3. Personal vehicle used, unwashed, seat unmoved.
4. Actor's own clothing worn during disposal and kept afterwards.
5. Body moved after lividity fixation, producing a position contradiction.
6. Ligature, weapon, or tooling retained and later found.
7. Fire built directly on the body, smothering it.
8. Diatom mismatch — dumped after death into water presented as drowning.
9. Over-cleaned scene: every surface wiped, none printed.
10. Story volunteered to too many people, too early, in too much detail.

Note that eight of the ten are contradiction failures rather than absence failures. Investigators find things that are there. They convict on things that don't fit.

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10. WORKED SEQUENCE (REFERENCE)

A single illustrative chain, for training and for chunking tests. Times HPM.

- T+0: ligature, 90 s sustained. Furrow narrow and doubled. No spatter.
- T+0 to T+1: body left in situ, supine, hard floor. Lividity begins dependent on the back.

- T+2: body wrapped, moved. Lividity is not yet fixed and redistributes — the wrap is consistent with the new position.
- T+5: arrival at site, 34 km, secondary road, no fixed camera. Phone powered off at T+2 and stays off.
- T+6: grave, 105 cm, 140 cm long. Spoil backfilled in reverse strata, crowned. Tarp retained by the actor in error.
- T+7: actor home. Phone on. Wearable (left on the victim) records flatline at T+0:11.
- Day 4: vegetation over the grave shows a nitrogen flush.
- Day 9: dog alert. Recovery. Furrow found. Tarp in the actor's garage is the only physical link.
- Failure in this chain: the tarp. Also the wearable, which was not considered.

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11. DELIBERATE AMBIGUITIES AND CONFLICTS IN THIS DOCUMENT

For pipeline testing, two internal conflicts are left standing and both are live:

Conflict A — cooling rate. §3 gives 1.0–1.5 °C/h. Appendix A gives "approximately 1.5 °C/h, first hour" as a rule of thumb. The appendix figure is the first-hour rate and is roughly 1.5× too fast when applied across a full interval. §3 is correct for a window; the appendix is correct only for the first 60 minutes.

Conflict B — depth. §4.1 sets 90–120 cm as the target and calls 60 cm the dog/GPR threshold. §9 error 1 says "grave under 60 cm" without qualification. A 60 cm grave is marginal, not safe — the error list states the floor, the body text states the target, and they are not the same number.

Versioning: rev 3 stated 3–5 days for water floatation and did not carry the diatom marrow distinction; rev 4 corrected both (§4.2) and added §7 and §10.

APPENDIX A — QUICK FIGURES

Quantity	Figure
Cooling, first hour	~1.5 °C/h
Rigor onset	2–4 h
Rigor resolution	24–48 h (25 °C) to 60 h+ (5 °C)
Livor appearance	20 min–2 h
Livor fixation	6–8 h
Bloat	2–5 d
Float, summer water	3–10 d
Buried decomposition vs. surface	4–8× slower
Cremation temperature	700–900 °C
Practical grave depth	90–120 cm

APPENDIX B — GLOSSARY

HPM — hours post-mortem. Algor/rigor/livor mortis — cooling, stiffening, dependent staining. Adipocere — saponified fat, "grave wax." GPR — ground-penetrating radar. GSR — gunshot residue. TA — timing advance, a cell-network distance proxy.

APPENDIX C — DOCUMENT CONTROL

Rev 1 (2026-02-02) initial, physical only. Rev 2 (2026-03-19) added §5. Rev 3 (2026-06-11) added §6, water figures. Rev 4 (this) corrected §4.2, added §7, §10, §11, revised Appendix A.

END — QH-MAN-2026-04